

TITLE:

IAPT Referral-to-Treatment (RTT) Analysis

Dataset Used:

nhstalkingtherapies_month_mar_2026_activity_performance.csv

SUBMITTED BY:

VALLURU SANDEEP KUMAR

DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Dataset Introduction

The IAPT (Improving Access to Psychological Therapies), now known as NHS Talking Therapies dataset, contains mental health service activity and performance information across NHS regions and providers in England. The dataset includes:

- Referral counts
- Waiting time measures
- Treatment access statistics
- Recovery indicators
- Performance metrics by organisation and service type

For this analysis, the focus is on Referral-to-Treatment (RTT) waiting time measures using:

- Mean waiting times
- Median waiting times
- Regional analysis
- Disorder/service category comparisons

The dataset file used:

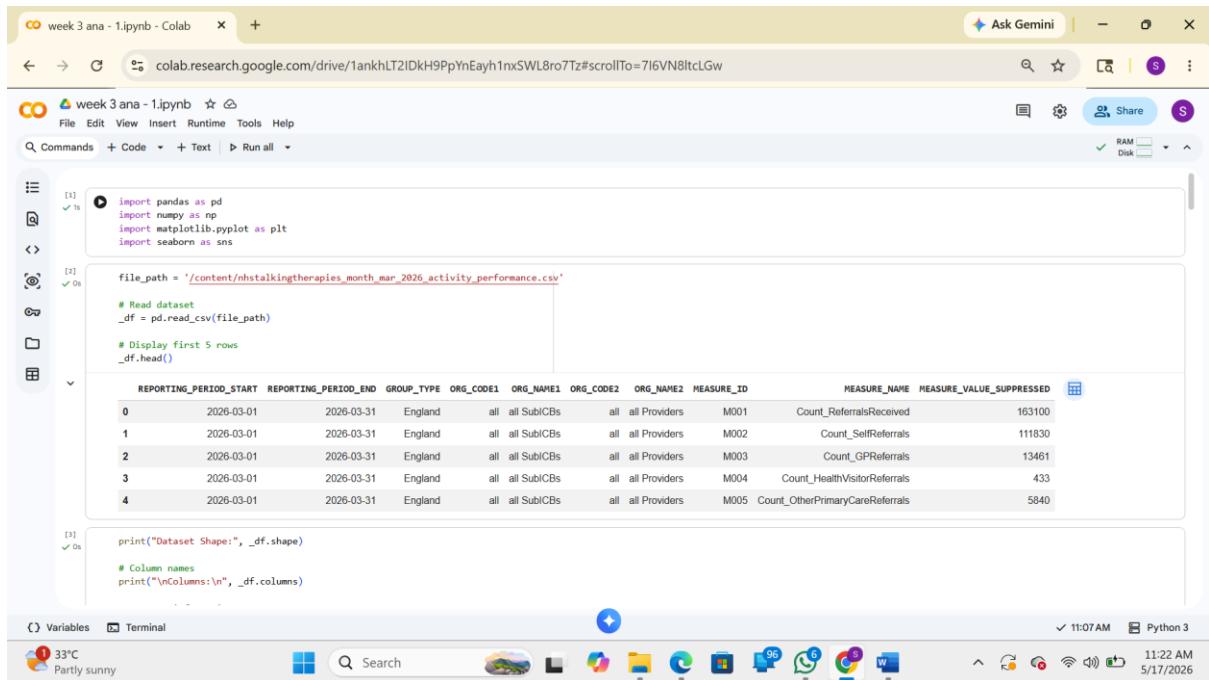
nhstalkingtherapies_month_mar_2026_activity_performance.csv

Technologies Used

Technology	Purpose
Python	Data analysis programming language
NumPy	Numerical operations and calculations
Pandas	Data cleaning and manipulation
Matplotlib	Data visualization and chart plotting
Seaborn	Statistical visualization and styling
Google Colab Notebook	Cloud-based Python execution environment

IAPT Referral-to-Treatment (RTT) Analysis

Importing libraries



The Colab notebook shows the following code cells:

```
[1] import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[2] file_path = '/content/nstalkingtherapies_month_mar_2026_activity_performance.csv'

# Read dataset
_df = pd.read_csv(file_path)

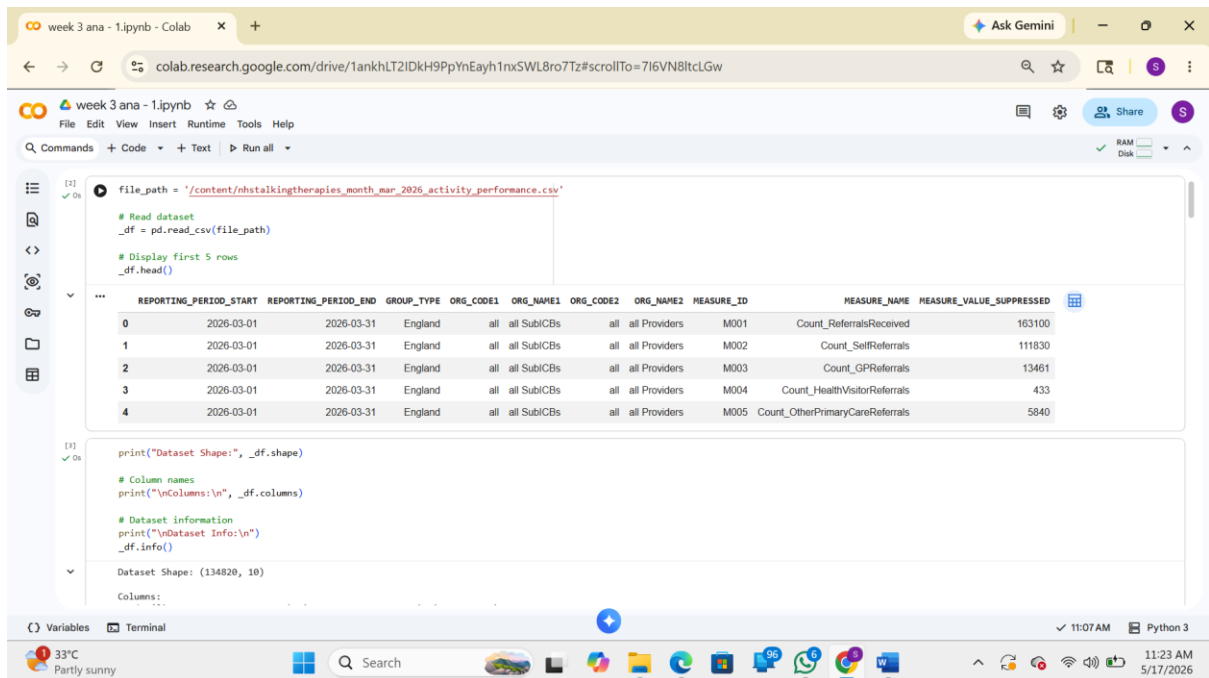
# Display first 5 rows
_df.head()
```

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
0	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M001	Count_ReferralsReceived	163100
1	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M002	Count_SelfReferrals	111830
2	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M003	Count_GPRReferrals	13461
3	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M004	Count_HealthVisitorReferrals	433
4	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M005	Count_OtherPrimaryCareReferrals	5840

```
[3] print("Dataset Shape:", _df.shape)

# Column names
print("\nColumns:\n", _df.columns)
```

Loading dataset



The Colab notebook shows the following code cells:

```
[2] file_path = '/content/nstalkingtherapies_month_mar_2026_activity_performance.csv'

# Read dataset
_df = pd.read_csv(file_path)

# Display first 5 rows
_df.head()
```

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
0	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M001	Count_ReferralsReceived	163100
1	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M002	Count_SelfReferrals	111830
2	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M003	Count_GPRReferrals	13461
3	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M004	Count_HealthVisitorReferrals	433
4	2026-03-01	2026-03-31	England	all	all SubICBs	all	all Providers	M005	Count_OtherPrimaryCareReferrals	5840

```
[3] print("Dataset Shape:", _df.shape)

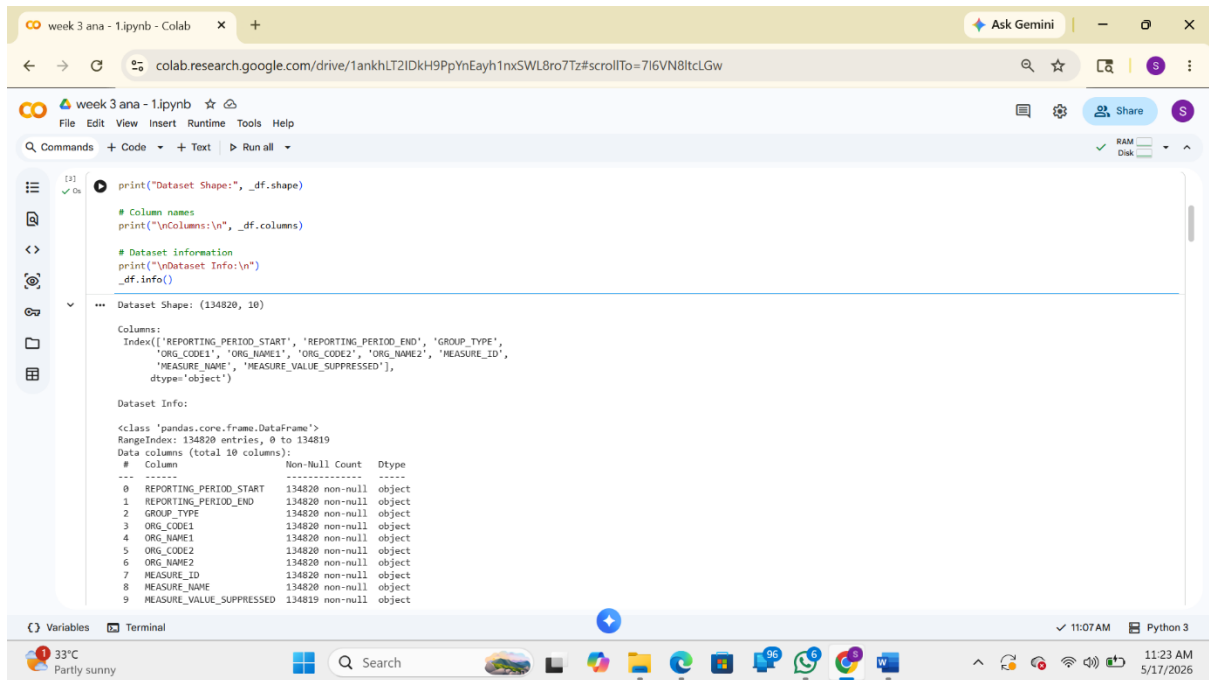
# Column names
print("\nColumns:\n", _df.columns)

# Dataset information
print("\nDataset Info:\n")
_df.info()
```

Dataset Shape: (134820, 10)

Columns:

Dataset information



The screenshot shows a Google Colab notebook interface. The code cell contains the following Python code:

```
[1]: print("Dataset Shape:", _df.shape)

# Column names
print("\nColumns:\n", _df.columns)

# Dataset information
print("\nDataset Info:\n")
_df.info()
```

The output of the code is displayed below the code cell:

```
Dataset Shape: (134820, 10)

Columns:
Index(['REPORTING_PERIOD_START', 'REPORTING_PERIOD_END', 'GROUP_TYPE',
      'ORG_CODE1', 'ORG_NAME1', 'ORG_CODE2', 'ORG_NAME2', 'MEASURE_ID',
      'MEASURE_NAME', 'MEASURE_VALUE_SUPPRESSED'],
      dtype='object')

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 134820 entries, 0 to 134819
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   REPORTING_PERIOD_START 134820 non-null object
 1   REPORTING_PERIOD_END   134820 non-null object
 2   GROUP_TYPE             134820 non-null object
 3   ORG_CODE1              134820 non-null object
 4   ORG_NAME1              134820 non-null object
 5   ORG_CODE2              134820 non-null object
 6   ORG_NAME2              134820 non-null object
 7   MEASURE_ID             134820 non-null object
 8   MEASURE_NAME           134820 non-null object
 9   MEASURE_VALUE_SUPPRESSED 134819 non-null object
```

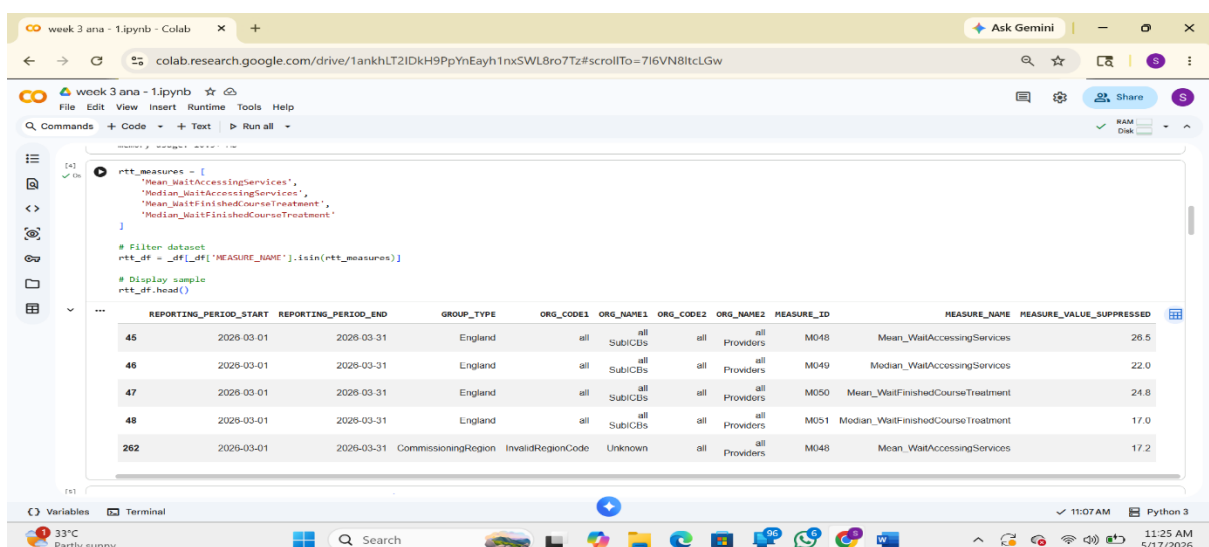
Rtt waiting time analysis

The dataset contains several waiting-time-related measures. In this analysis, we focus on:

- Mean_WaitAccessingServices
- Median_WaitAccessingServices
- Mean_WaitFinishedCourseTreatment
- Median_WaitFinishedCourseTreatment

These metrics help identify average and median waiting times experienced by patients.

Filter RTT Measures



The screenshot shows a Google Colab notebook interface. The code cell contains the following Python code:

```
[4]: rtt_measures = [
    "Mean_WaitAccessingServices",
    "Median_WaitAccessingServices",
    "Mean_WaitFinishedCourseTreatment",
    "Median_WaitFinishedCourseTreatment"
]

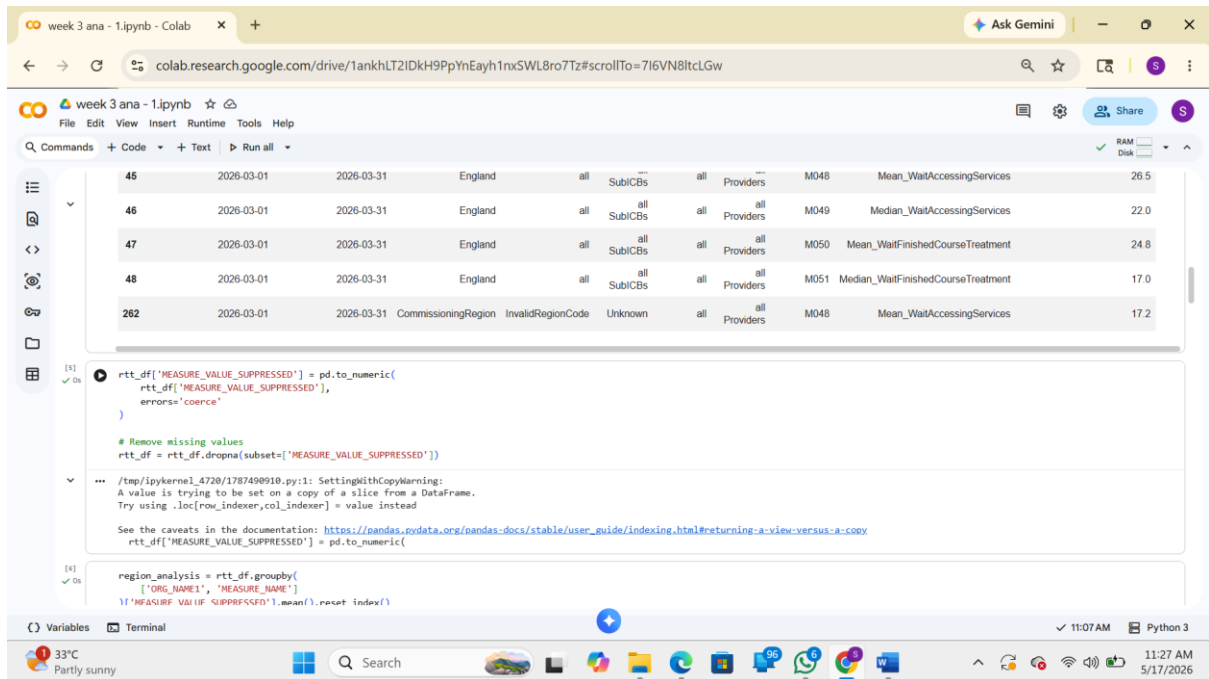
# Filter dataset
rtt_df = _df[_df['MEASURE_NAME'].isin(rtt_measures)]

# Display sample
rtt_df.head()
```

The output of the code is displayed below the code cell, showing a sample of the filtered dataset:

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
45	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M048	Mean_WaitAccessingServices	26.5
46	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M049	Median_WaitAccessingServices	22.0
47	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M050	Mean_WaitFinishedCourseTreatment	24.8
48	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M051	Median_WaitFinishedCourseTreatment	17.0
262	2026-03-01	2026-03-31	CommissioningRegion	InvalidRegionCode	Unknown	all	Providers	M048	Mean_WaitAccessingServices	17.2

Convert Measure Values to Numeric



The screenshot shows a Google Colab notebook with the following code and output:

```
[9]: rtt_df["MEASURE_VALUE_SUPPRESSED"] = pd.to_numeric(
    rtt_df["MEASURE_VALUE_SUPPRESSED"],
    errors='coerce'
)

# Remove missing values
rtt_df = rtt_df.dropna(subset=["MEASURE_VALUE_SUPPRESSED"])

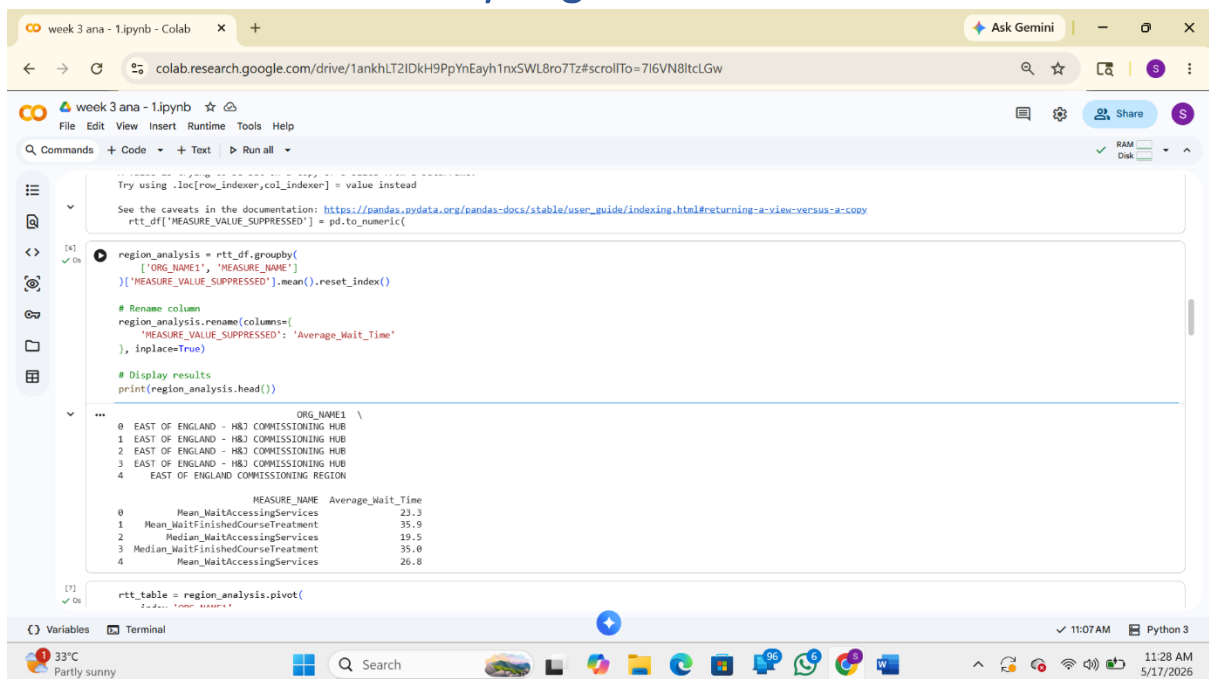
... /tmp/ipykernel_4720/1787490910.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
rtt_df["MEASURE_VALUE_SUPPRESSED"] = pd.to_numeric(

[10]: region_analysis = rtt_df.groupby(
    ["ORG_NAME1", "MEASURE_NAME"]
    )[["MEASURE_VALUE_SUPPRESSED"]].mean().reset_index()
```

	ORG_NAME1	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED							
45	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M048	Mean_WaitAccessingServices	26.5
46	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M049	Median_WaitAccessingServices	22.0
47	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M050	Mean_WaitFinishedCourseTreatment	24.8
48	2026-03-01	2026-03-31	England	all	SubICBs	all	Providers	M051	Median_WaitFinishedCourseTreatment	17.0
262	2026-03-01	2026-03-31	CommissioningRegion	InvalidRegionCode	Unknown	all	Providers	M048	Mean_WaitAccessingServices	17.2

Mean and Median RTT by Region



The screenshot shows a Google Colab notebook with the following code and output:

```
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
rtt_df["MEASURE_VALUE_SUPPRESSED"] = pd.to_numeric(

[10]: region_analysis = rtt_df.groupby(
    ["ORG_NAME1", "MEASURE_NAME"]
    )[["MEASURE_VALUE_SUPPRESSED"]].mean().reset_index()

# Rename column
region_analysis.rename(columns={
    "MEASURE_VALUE_SUPPRESSED": "Average_Wait_Time"
}, inplace=True)

# Display results
print(region_analysis.head())

... 0 EAST OF ENGLAND - H&J COMMISSIONING HUB
1 EAST OF ENGLAND - H&J COMMISSIONING HUB
2 EAST OF ENGLAND - H&J COMMISSIONING HUB
3 EAST OF ENGLAND - H&J COMMISSIONING HUB
4 EAST OF ENGLAND COMMISSIONING REGION

MEASURE_NAME Average_Wait_Time
0 Mean_WaitAccessingServices 23.3
1 Mean_WaitFinishedCourseTreatment 35.9
2 Median_WaitAccessingServices 19.5
3 Median_WaitFinishedCourseTreatment 35.0
4 Mean_WaitAccessingServices 26.8

[11]: rtt_table = region_analysis.pivot(
    index="ORG_NAME1",
    columns="MEASURE_NAME",
    values="Average_Wait_Time")
```

ORG_NAME1 \ MEASURE_NAME	Mean_WaitAccessingServices	Mean_WaitFinishedCourseTreatment	Median_WaitAccessingServices	Median_WaitFinishedCourseTreatment
0 EAST OF ENGLAND - H&J COMMISSIONING HUB	23.3	35.9	19.5	35.0
1 EAST OF ENGLAND - H&J COMMISSIONING HUB	23.3	35.9	19.5	35.0
2 EAST OF ENGLAND - H&J COMMISSIONING HUB	23.3	35.9	19.5	35.0
3 EAST OF ENGLAND - H&J COMMISSIONING HUB	23.3	35.9	19.5	35.0
4 EAST OF ENGLAND COMMISSIONING REGION	26.8	35.0	19.5	35.0

RTT Analysis Table

week 3 ana - 1.ipynb - Colab

Ask Gemini

colab.research.google.com/drive/1ankhLT2IDkH9PpYnEayh1nxSWL8ro7Tz#scrollTo=7I6VN8ltcLGw

File Edit View Insert Runtime Tools Help

Commands + Code + Text + Run all

RAM Disk

[7] ✓ On

```
rtt_table = region_analysis.pivot(
    index='ORG_NAME1',
    columns='MEASURE_NAME',
    values='Average_Wait_Time'
)

# Display table
print(rtt_table)
```

MEASURE_NAME	Mean_WaitAccessingServices \
ORG_NAME1	
EAST OF ENGLAND - H&J COMMISSIONING HUB	23.300000
EAST OF ENGLAND COMMISSIONING REGION	26.800000
LONDON COMMISSIONING REGION	25.500000
MIDLANDS COMMISSIONING REGION	30.000000
NATIONAL COMMISSIONING HUB 1	17.600000
...	...
NORTH WEST COMMISSIONING REGION	30.200000
SOUTH EAST COMMISSIONING REGION	21.000000
SOUTH WEST COMMISSIONING REGION	29.000000
Unknown	17.200000
all SubICBs	31.239231

MEASURE_NAME	Mean_WaitFinishedCourseTreatment \
ORG_NAME1	
EAST OF ENGLAND - H&J COMMISSIONING HUB	35.900000
EAST OF ENGLAND COMMISSIONING REGION	26.700000
LONDON COMMISSIONING REGION	23.200000
MIDLANDS COMMISSIONING REGION	28.600000
NATIONAL COMMISSIONING HUB 1	14.900000
...	...
NORTH WEST COMMISSIONING REGION	30.400000
SOUTH EAST COMMISSIONING REGION	18.200000
SOUTH WEST COMMISSIONING REGION	21.000000

Variables Terminal

11:07 AM Python 3

33°C Partly sunny

Search

11:29 AM 5/17/2026

week 3 ana - 1.ipynb - Colab

Ask Gemini

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File Edit View Insert Runtime Tools Help

Commands + Code + Text + Run all

RAM Disk

[117] rows x 4 columns

```
SOUTH EAST COMMISSIONING REGION 18.200000
SOUTH WEST COMMISSIONING REGION 23.000000
Unknown 17.400000
all SubICBs 33.68862

MEASURE_NAME Median_WaitAccessingServices \
ORG_NAME1
EAST OF ENGLAND - H&J COMMISSIONING HUB 19.500000
EAST OF ENGLAND COMMISSIONING REGION 26.000000
LONDON COMMISSIONING REGION 21.000000
MIDLANDS COMMISSIONING REGION 22.000000
NATIONAL COMMISSIONING HUB 1 14.000000
...
NORTH WEST COMMISSIONING REGION 27.000000
SOUTH EAST COMMISSIONING REGION 19.000000
SOUTH WEST COMMISSIONING REGION 28.000000
Unknown 18.000000
all SubICBs 26.369231

MEASURE_NAME Median_WaitFinishedCourseTreatment
ORG_NAME1
EAST OF ENGLAND - H&J COMMISSIONING HUB 35.000000
EAST OF ENGLAND COMMISSIONING REGION 17.000000
LONDON COMMISSIONING REGION 15.000000
MIDLANDS COMMISSIONING REGION 19.000000
NATIONAL COMMISSIONING HUB 1 13.000000
...
NORTH WEST COMMISSIONING REGION 22.000000
SOUTH EAST COMMISSIONING REGION 14.000000
SOUTH WEST COMMISSIONING REGION 19.000000
Unknown 13.000000
all SubICBs 27.976744
```

Variables Terminal

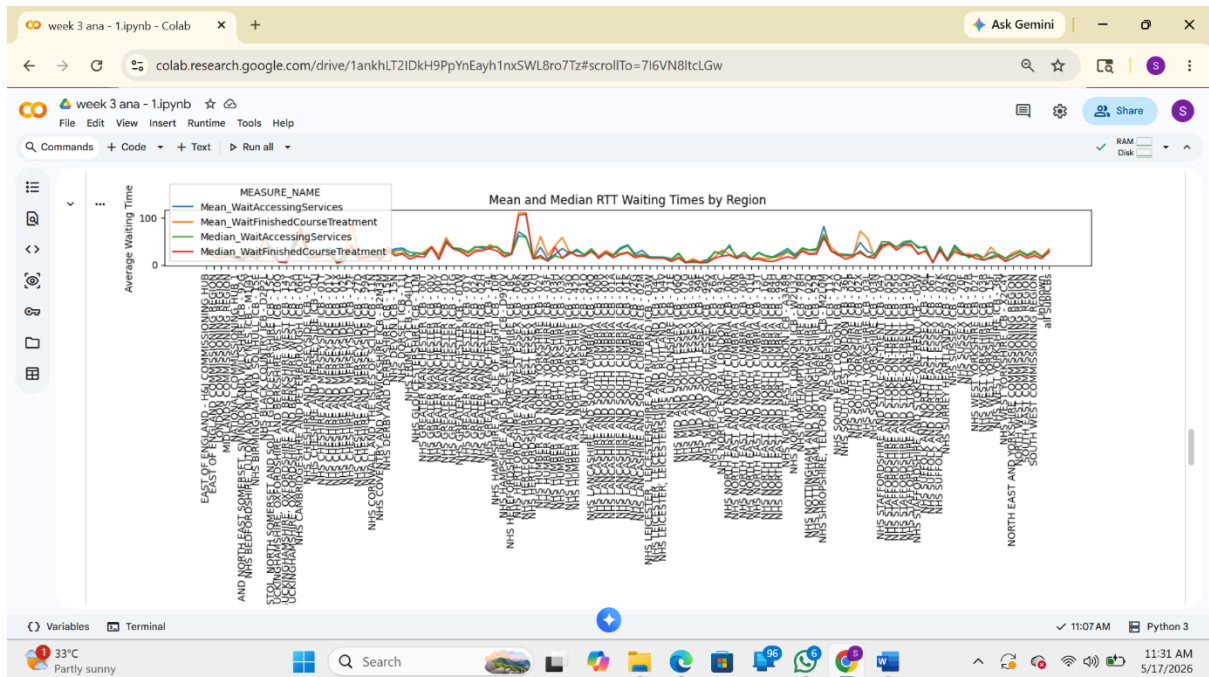
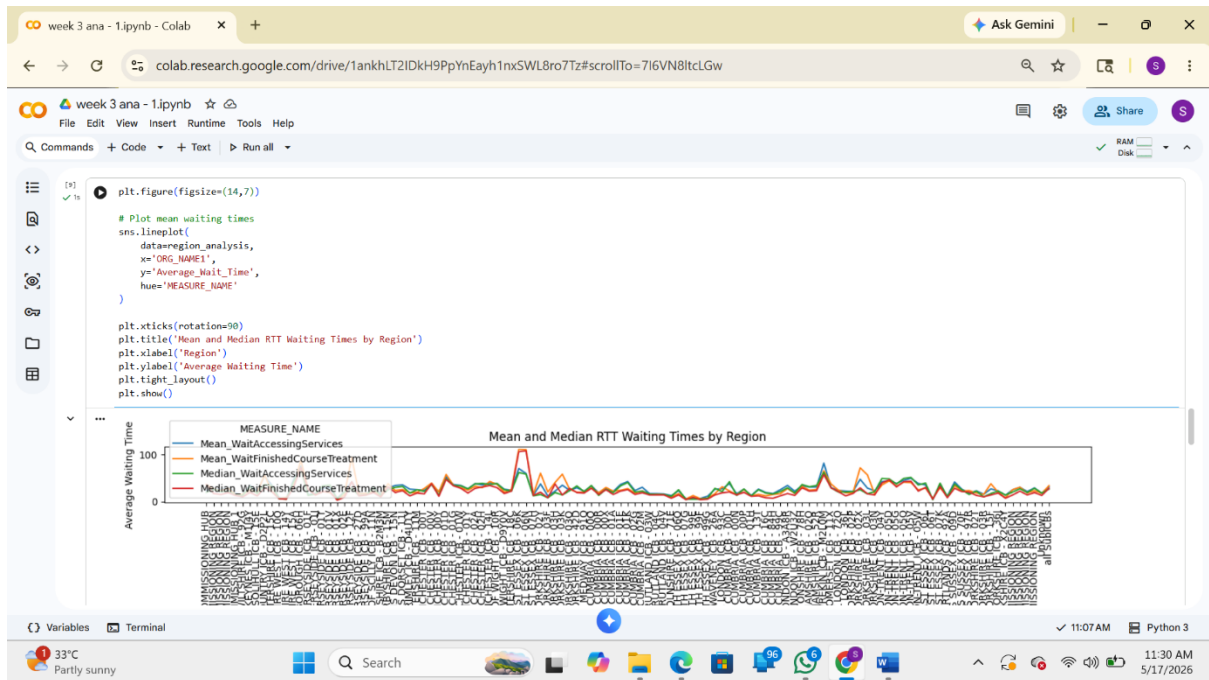
11:07 AM Python 3

33°C Partly sunny

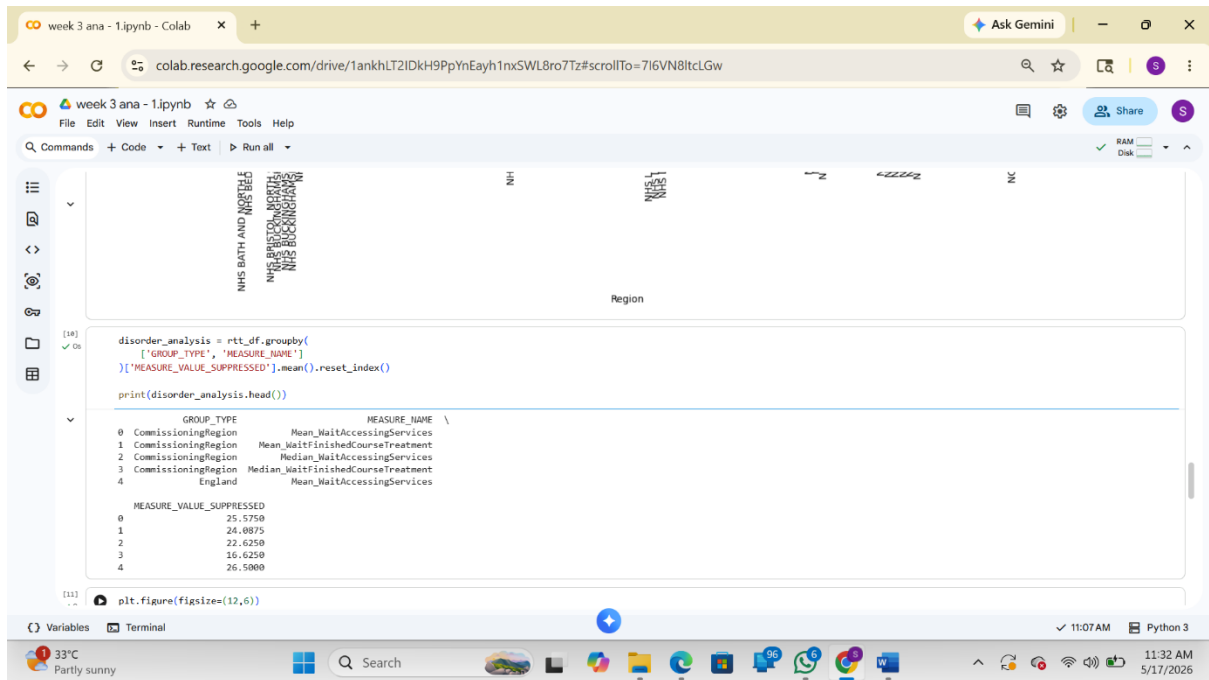
Search

11:29 AM 5/17/2026

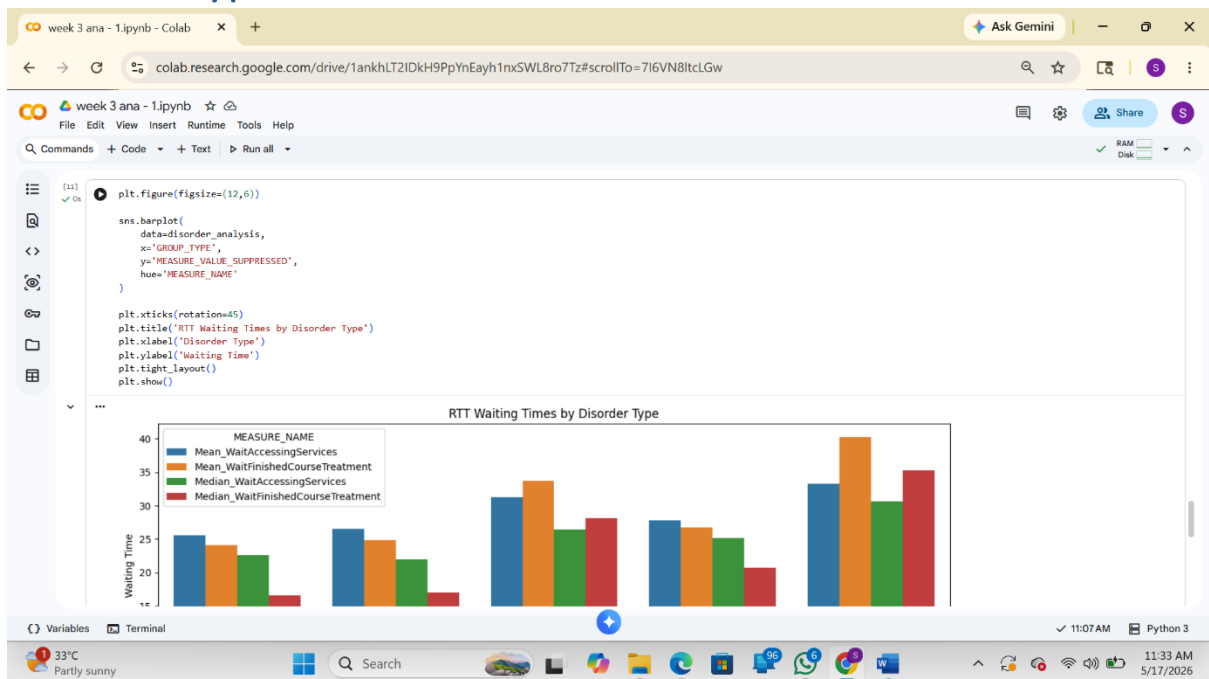
RTT Analysis Chart using Matplotlib

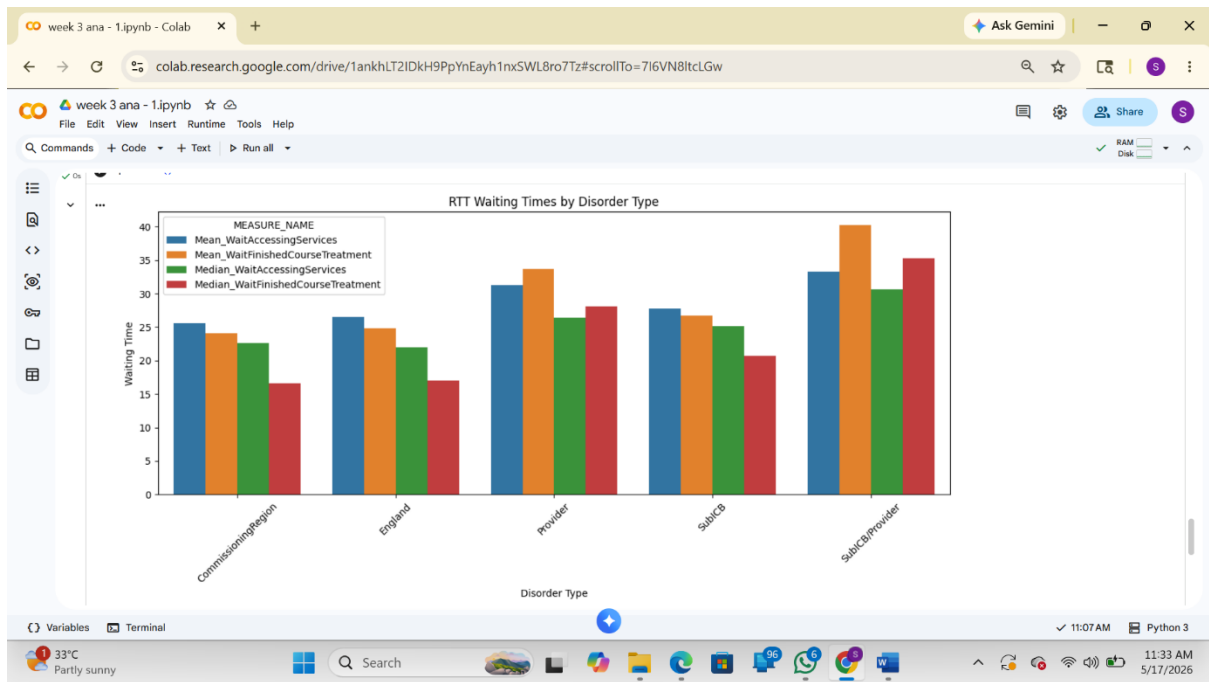


Disorder Type Analysis

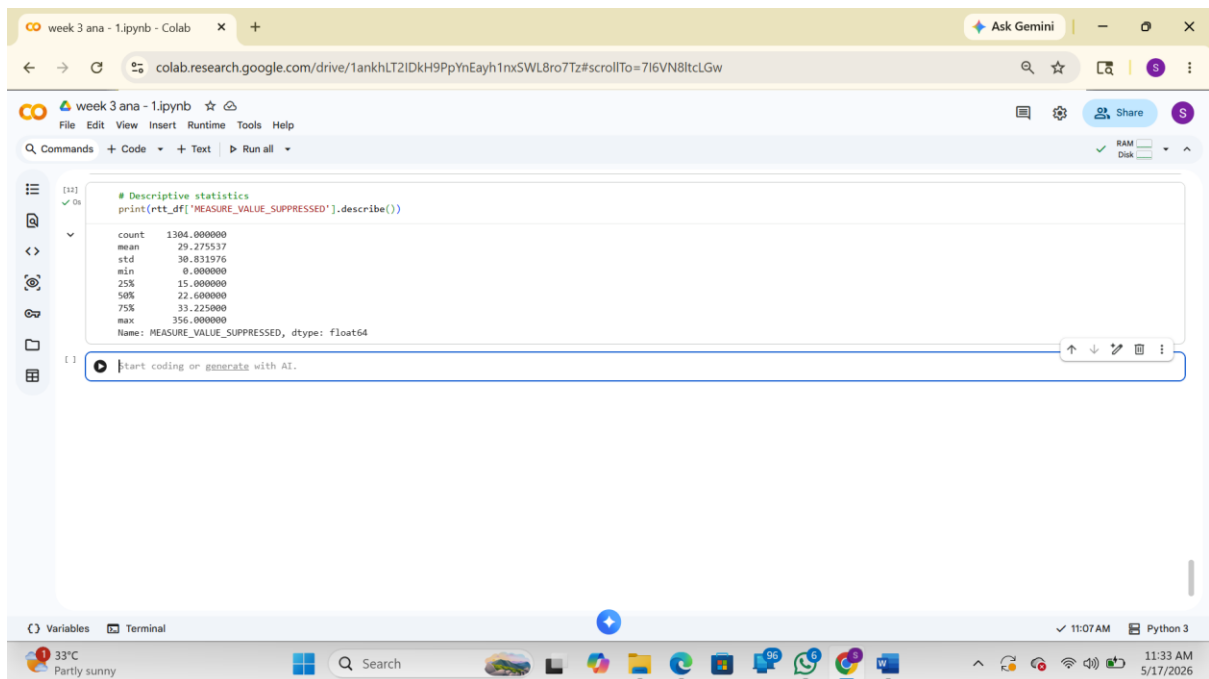


Disorder Type Chart





Statistical Summary



Key Insights

- Mean waiting time measures provide the average delay experienced by patients.
- Median waiting time measures reduce the impact of extreme outliers.
- Regional analysis helps identify areas with longer treatment delays.

- Disorder-level analysis helps understand which service groups require improved access.
- Visualization makes it easier to compare NHS regions and treatment performance.

Conclusion

This IAPT RTT analysis provides insights into NHS Talking Therapies waiting times across regions and service categories. Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset was cleaned, analysed, and visualised effectively.

The analysis helps:

- Monitor NHS service performance
- Identify long waiting-time regions
- Compare mean and median RTT measures
- Support healthcare decision-making
- Improve patient access to mental health services

Google Colab provides an easy and efficient environment for executing the analysis and generating visual reports.